

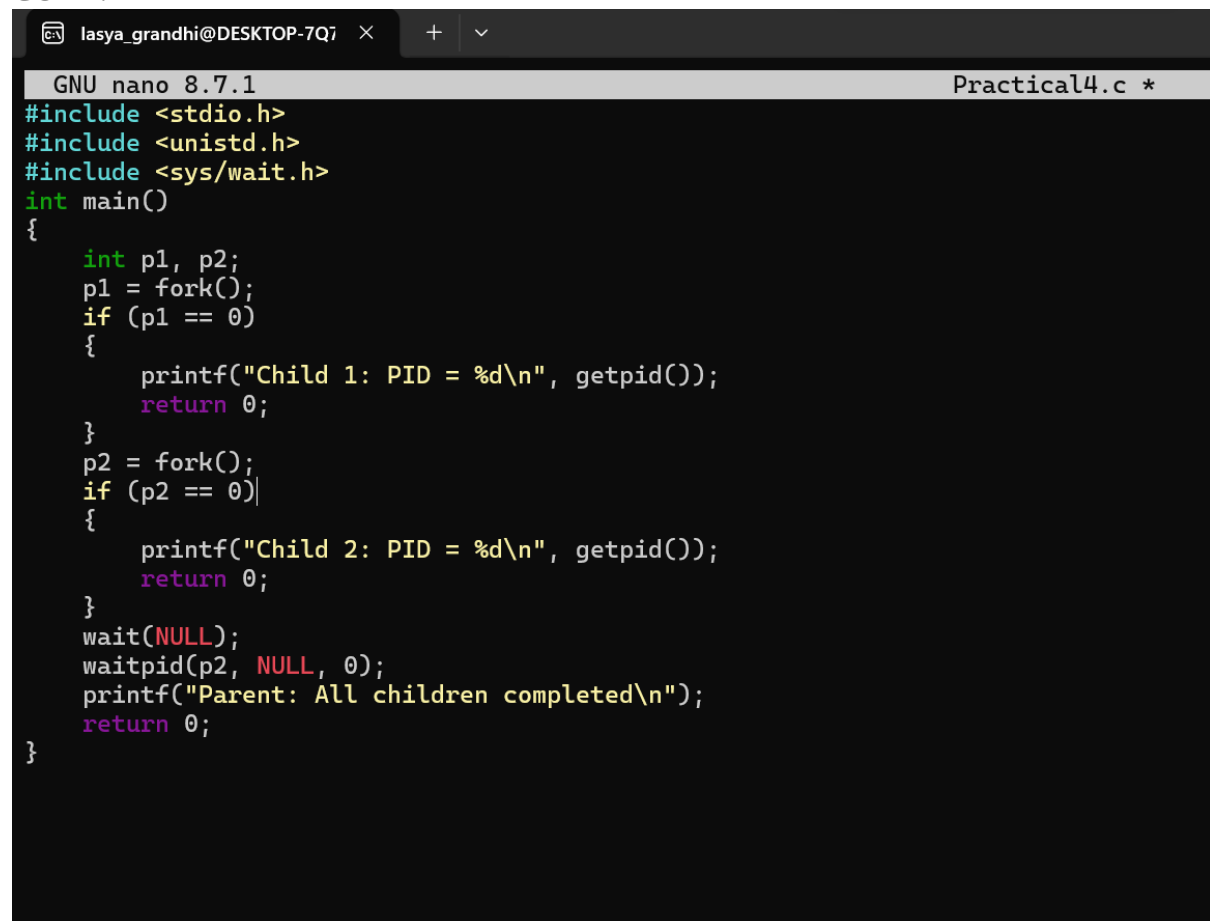
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4. Write a C program where a parent process creates multiple child processes and synchronizes their completion using wait() and waitpid(). Compare the behavior of both functions.

Create a scenario where a child process becomes a zombie process. Investigate the process table and then modify the program to eliminate zombie processes using proper synchronization techniques.

TASK1

CODE:

A screenshot of a code editor window titled 'lasya_grandhi@DESKTOP-7Q7' with a tab for 'Practical4.c *'. The editor shows the GNU nano 8.7.1 interface with a C program. The code includes <stdio.h>, <unistd.h>, and <sys/wait.h>. The main function forks two child processes. The first child prints its PID and returns 0. The second child prints its PID and returns 0. The parent process then calls wait(NULL) and waitpid(p2, NULL, 0) before printing 'Parent: All children completed\n' and returning 0.

```
GNU nano 8.7.1 Practical4.c *
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
int main()
{
    int p1, p2;
    p1 = fork();
    if (p1 == 0)
    {
        printf("Child 1: PID = %d\n", getpid());
        return 0;
    }
    p2 = fork();
    if (p2 == 0)
    {
        printf("Child 2: PID = %d\n", getpid());
        return 0;
    }
    wait(NULL);
    waitpid(p2, NULL, 0);
    printf("Parent: All children completed\n");
    return 0;
}
```

OUTPUT:

```
lasya_grandhi@DESKTOP-7Q7B8KJ:~$ nano Practical4.c
lasya_grandhi@DESKTOP-7Q7B8KJ:~$ gcc Practical4.c
lasya_grandhi@DESKTOP-7Q7B8KJ:~$ ./a.out
Child 1: PID = 742
Child 2: PID = 743
Parent: All children completed
lasya_grandhi@DESKTOP-7Q7B8KJ:~$
```

TASK2:

CODE:

```
GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
int main(){
    int pid;
    pid = fork();
    if (pid == 0){
        printf("Child process completed\n");
        return 0;
    }else{
        sleep(5);
        wait(NULL);
        printf("Parent collected child\n");
    }
    return 0;
}
```

OUTPUT:

```
lasya_grandhi@DESKTOP-7Q7B8KJ:~$ nano Practical4b.c
lasya_grandhi@DESKTOP-7Q7B8KJ:~$ gcc Practical4b.c
lasya_grandhi@DESKTOP-7Q7B8KJ:~$ ./a.out
Child process completed
Parent collected child
lasya_grandhi@DESKTOP-7Q7B8KJ:~$
```